

Logic

II. First-order logic and Peano–Heyting arithmetic

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A richer structure of propositions

When talking about mathematical structures such as Peano–Heyting (natural number) arithmetic, we use statements such as

for every x , if $x \neq 0$ then **there exists** y such that $\text{suc } y = x$ that involve *first-order quantification* over individual values in a *domain*.

(The function suc is the *successor* function on natural numbers.)

This motivates us to extend propositional logic with first-order quantification, and the result is called *first-order logic*.

Going from propositional logic to first-order logic requires more than enriching the language with quantification though.

Substitution

Variables are to be *substituted* for. For example, from

for every x , if $x \neq 0$ then there exists y such that $\text{succ } y = x$

we should be able to deduce

if $1 \neq 0$ then there exists y such that $\text{succ } y = 1$

by substituting 1 for the variable x .

The structure of (previously) atomic propositions must be refined so the variable x can be substituted.

Sub-atomic structure

Syntactically, the proposition $suc\ y = x$ may be analysed as follows:

- '=' is a *predicate symbol* that accepts two *terms*, and
- 'suc' is a *function symbol* that can be used to construct more complex terms, which can contain variables.

Each symbol has an associated natural number called its *arity*, which specifies the number of sub-terms the symbol expects.

Terms

Let $\mathcal{IV} := \{x, y, z, \dots\}$ be an infinite set of individual variables.

Definition. Given a set $\mathcal{F}$ of symbols with arities, the set $\text{TERM}_{\mathcal{F}}$ of *terms* is inductively defined by the following rules:

- $v : \text{TERM}_{\mathcal{F}}$ for all $v : \mathcal{IV}$;
- $f t_1 \dots t_n : \text{TERM}_{\mathcal{F}}$ for all $f/n : \mathcal{F}$ (function symbol f with arity n) and $t_1, \dots, t_n : \text{TERM}_{\mathcal{F}}$.

Example. For terms in Peano–Heyting arithmetic, we choose $\mathcal{F} := \{\text{zero}/0, \text{suc}/1, \text{add}/2, \text{mul}/2\}$.

Question. Why does $\mathcal{IV}$ need to be infinite?

First-order formulas

Definition. A *signature* $\mathcal{S}$ is a pair of sets $(\mathcal{P}, \mathcal{F})$ of symbols with arities, where elements of $\mathcal{P}$ are called *predicate symbols* and elements of $\mathcal{F}$ are called *function symbols*.

Definition. Given a signature $\mathcal{S} = (\mathcal{P}, \mathcal{F})$, the set $\text{FORM}_{\mathcal{S}}$ of *first-order formulas* is inductively defined by the following rules:

- $\perp : \text{FORM}_{\mathcal{S}}$;
- $p \ t_1 \dots t_n : \text{FORM}_{\mathcal{S}}$ for all $p/n : \mathcal{P}$ and $t_1, \dots, t_n : \text{TERM}_{\mathcal{F}}$;
- $\varphi \wedge \psi : \text{FORM}_{\mathcal{S}}$ for all $\varphi, \psi : \text{FORM}_{\mathcal{S}}$;
- $\varphi \vee \psi : \text{FORM}_{\mathcal{S}}$ for all $\varphi, \psi : \text{FORM}_{\mathcal{S}}$;
- $\varphi \rightarrow \psi : \text{FORM}_{\mathcal{S}}$ for all $\varphi, \psi : \text{FORM}_{\mathcal{S}}$;
- $\forall v. \varphi : \text{FORM}_{\mathcal{S}}$ for all $v : \mathcal{IV}$ and $\varphi : \text{FORM}_{\mathcal{S}}$;
- $\exists v. \varphi : \text{FORM}_{\mathcal{S}}$ for all $v : \mathcal{IV}$ and $\varphi : \text{FORM}_{\mathcal{S}}$.

Instantiations of first-order logic

Example. The signature for Peano–Heyting arithmetic consists of $\mathcal{P} := \{ \text{Eq}/2 \}$ and $\mathcal{F} := \{ \text{zero}/0, \text{suc}/1, \text{add}/2, \text{mul}/2 \}$.

The proposition

for every x , if $x \neq 0$ then **there exists** y such that $\text{suc } y = x$

is written formally as

$$\forall x. ((\neg \text{Eq } x \text{ zero}) \rightarrow \exists y. \text{Eq } (\text{suc } y) x)$$

Question. Why these particular parentheses? Why not more?
Why not less?

Instantiations of first-order logic (continued)

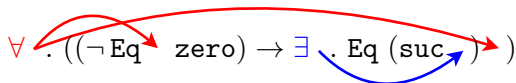
Example. For Tarski's axioms for Euclidean geometry, individuals are points, and the signature consists of $\mathcal{P} := \{ \text{Eq}/2, \text{B}/3, \text{Ed}/4 \}$ and $\mathcal{F} := \{ \}$, where Eq is equality of points, $\text{B } x \ y \ z$ means y is between x and z , and $\text{Ed } x \ y \ z \ w$ means the distances from x to y and from z to w are equal.

Exercise. Formalise the 'axiom of segment construction': given any segment $\overline{ab}$, any segment $\overline{xy}$ can be extended to a segment $\overline{xz}$ such that the segment $\overline{yz}$ is as long as $\overline{ab}$.

Representations of binding (intuitively)

The choice of variable in *binders* $\forall v$ and $\exists v$ is not important and can be changed by *α -conversion*; what is important is the *binding* structure, that is, when a binder is instantiated, which variable occurrences are to be substituted for.

‘Ideally’ we would want to represent binding graphically, for example writing/drawing



which can be represented textually as

$$\forall x. ((\neg \text{Eq } x \text{ zero}) \rightarrow \exists y. \text{Eq } (\text{succ } y) x)$$

$$\forall z. ((\neg \text{Eq } z \text{ zero}) \rightarrow \exists y. \text{Eq } (\text{succ } y) z)$$

and so on (these are called *α -equivalent* formulas), but not as

$$\forall x. ((\neg \text{Eq } x \text{ zero}) \rightarrow \exists x. \text{Eq } (\text{succ } x) x)$$

which represents a different binding structure.

Substitution and variable capture (informally)

A variable occurrence under a binder of the same variable is *bound*; otherwise it is *free*.

Write $\varphi[t/v]$ for the result of substituting the term t for the free occurrences of the variable v in the formula φ .

Naive substitution can result in *variable capture*, which creates unintended variable binding. For example:

$$\begin{aligned} & ((\neg \text{Eq } x \text{ zero}) \rightarrow \exists y. \text{Eq } (\text{suc } y) x) [\text{add } z \ y/x] \\ & \neq (\neg \text{Eq } (\text{add } z \ y) \text{ zero}) \rightarrow \exists y. \text{Eq } (\text{suc } y) (\text{add } z \ y) \end{aligned}$$

Textual substitution

Substitution $\varphi[t/v]$ can be performed directly by replacing every occurrence of v in φ with t if a side condition is satisfied:

all the binder variables in φ are different from v and the (free) variables in t .

And the side condition can always be satisfied by α -converting φ suitably.

Exercise. Let $\varphi := (\exists z. \text{Eq } x \text{ (add } z \text{ y)}) \wedge \exists y. \text{Eq } x \text{ (add } y \text{ y)}$.

- Which variable occurrences are free in φ ?
- Compute $\varphi[\text{zero}/y]$.
- Compute $\varphi[\text{suc } z/x]$.

Question. Why is the side condition sufficient?

Formal definitions of free variables

Definition. Let $\mathcal{F}$ be a set of function symbols with arities. The set of (free) variables in a term is defined by

$$\begin{aligned} FV: \text{TERM}_{\mathcal{F}} &\rightarrow \wp \mathcal{IV} \\ FV v &= \{v\} && \text{for } v: \mathcal{IV} \\ FV (f t_1 \dots t_n) &= FV t_1 \cup \dots \cup FV t_n && \text{for } f/n: \mathcal{F} \end{aligned}$$

Definition. Let $\mathcal{S} = (\mathcal{P}, \mathcal{F})$ be a signature. The set of free variables in a first-order formula is defined by

$$\begin{aligned} FV: \text{FORM}_{\mathcal{S}} &\rightarrow \wp \mathcal{IV} \\ FV \perp &= \{\} \\ FV (p t_1 \dots t_n) &= FV t_1 \cup \dots \cup FV t_n && \text{for } p/n: \mathcal{P} \\ FV (\varphi \bullet \psi) &= FV \varphi \cup FV \psi && \text{where } \bullet = \wedge, \vee, \rightarrow \\ FV (\forall v. \varphi) &= FV \varphi \setminus \{v\} \\ FV (\exists v. \varphi) &= FV \varphi \setminus \{v\} \end{aligned}$$

Exercise. Define α -conversion, substitution, etc formally.

Intuitionistic meaning of quantifiers

We assume a set $\mathcal{D}$, called the *domain (of discourse)*, over which we quantify.

A formula φ with free variable v describes a property of an element in $\mathcal{D}$ referred to as v .

- A proof of $\forall v. \varphi$ is a method that, for every element $d : \mathcal{D}$, produces a proof of φ for d .
- A proof of $\exists v. \varphi$ is an element $d : \mathcal{D}$ —called the *witness*— and a proof of φ for d .

To obtain a deduction system for intuitionistic first-order logic, we extend NJ with introduction and elimination rules for ' $\forall$ ' and ' $\exists$ '.

Introducing and eliminating '∀'

$$\frac{\Gamma \vdash \varphi}{\Gamma \vdash \forall v. \varphi} (\forall I) \qquad \frac{\Gamma \vdash \forall v. \varphi}{\Gamma \vdash \varphi [t/v]} (\forall E)$$

(∀I) has a side condition that $v \notin FV \Gamma$, where

$$FV \Gamma := \bigcup_{\psi \in \Gamma} FV \psi$$

That is, to generalise φ to $\forall v. \varphi$, there should be no assumptions about v .

Exercise. Derive

$$\vdash (\forall x. \forall y. P \ x \ y) \rightarrow \forall x. \forall y. P \ y \ x$$

Exercise. Try to derive

$$\vdash \forall x. (Eq \ x \ zero \rightarrow \forall x. Eq \ x \ zero)$$

Introducing and eliminating '∃'

$$\frac{\Gamma \vdash \varphi[t/v]}{\Gamma \vdash \exists v. \varphi} (\exists I) \qquad \frac{\Gamma \vdash \exists v. \varphi \quad \Gamma, \varphi \vdash \psi}{\Gamma \vdash \psi} (\exists E)$$

(∃E) has a side condition that $v \notin FV \Gamma \cup FV \psi$.

That is, v should be a name chosen freshly for the witness.

Exercise. Derive

$$\vdash (\exists x. \forall y. P \ x \ y) \rightarrow \forall y. \exists x. P \ x \ y$$

Exercise. Try to finish deriving

$$\frac{\frac{\exists x. P \ x \vdash P \ x}{\exists x. P \ x \vdash \forall x. P \ x} (\forall I)}{\vdash (\exists x. P \ x) \rightarrow \forall x. P \ x} (\rightarrow I)$$

Exercise. Try to derive

$$\vdash \forall x. ((\neg \text{Eq } x \ \text{zero}) \rightarrow \neg \exists x. \text{Eq } x \ \text{zero})$$

Remark on intuitionistic existential quantification

We can derive

$$\vdash (\exists v. \varphi) \rightarrow (\neg \forall v. \neg \varphi) \quad \text{but not} \quad \vdash (\neg \forall v. \neg \varphi) \rightarrow (\exists v. \varphi)$$

Intuitionistic existential quantification is stronger than its classical counterpart (which is definable from ' $\forall$ ' and ' $\neg$ ').

Exercise. Derive $\vdash (\neg \forall v. \neg \varphi) \rightarrow (\exists v. \varphi)$ assuming the law of excluded middle or the principle of indirect proof.

Similar to Glivenko's theorem for propositional logic, there are double-negation translations for embedding classical first-order logic into intuitionistic first-order logic.

Peano–Heyting arithmetic

The signature for Peano–Heyting arithmetic consists of $\mathcal{P} := \{ \text{Eq}/2 \}$ and $\mathcal{F} := \{ \text{zero}/0, \text{suc}/1, \text{add}/2, \text{mul}/2 \}$.

In the following we write $t_1 \equiv t_2$ for $\text{Eq } t_1 \ t_2$, $t_1 + t_2$ for $\text{add } t_1 \ t_2$, and $t_1 \times t_2$ for $\text{mul } t_1 \ t_2$.

Properties about these symbols are postulated by the (infinite) list **PA** of *Peano axioms*, to be given in the next two slides.

Definition. A formula φ is a *sentence* exactly when $FV \varphi = \emptyset$.

Definition. A list of sentences is called a *theory*, whose elements are called *axioms*.

Definition. A sentence derivable from a theory $\mathcal{T}$ is called a *theorem* of $\mathcal{T}$.

Example. **PA** is a theory.

Question. Why include only sentences in a theory?

Peano axioms: predicate and function symbols

The first three axioms make Eq an equivalence relation.

$$\text{reflexivity} \quad := \quad \forall x. x \equiv x$$

$$\text{transitivity} \quad := \quad \forall x. \forall y. \forall z. ((x \equiv y \wedge y \equiv z) \rightarrow x \equiv z)$$

$$\text{symmetry} \quad := \quad \forall x. \forall y. (x \equiv y \rightarrow y \equiv x)$$

The next three axioms are about zero and suc.

$$\text{disjointness} \quad := \quad \forall x. \neg (\text{suc } x \equiv \text{zero})$$

$$\text{injectivity} \quad := \quad \forall x. \forall y. (\text{suc } x \equiv \text{suc } y \rightarrow x \equiv y)$$

$$\text{congruence} \quad := \quad \forall x. \forall y. (x \equiv y \rightarrow \text{suc } x \equiv \text{suc } y)$$

The following four axioms characterise add and mul.

$$\text{additionZ} \quad := \quad \forall y. \text{zero} + y \equiv y$$

$$\text{additionS} \quad := \quad \forall x. \forall y. (\text{suc } x) + y \equiv \text{suc } (x + y)$$

$$\text{multiplicationZ} \quad := \quad \forall y. \text{zero} \times y \equiv \text{zero}$$

$$\text{multiplicationS} \quad := \quad \forall x. \forall y. (\text{suc } x) \times y \equiv y + (x \times y)$$

Peano axioms: induction

Finally there is an *axiom scheme* that generates infinite instances of the induction principle on natural numbers: for every formula φ (called the *motive*) and variable v there is an axiom

$$\text{induction}_{\varphi, v} := \\ \text{closure} ((\varphi [\text{zero}/v] \wedge \forall v. (\varphi \rightarrow (\varphi [\text{suc } v/v]))) \rightarrow \forall v. \varphi)$$

Definition. The *universal closure* of a formula ψ is defined by

$$\text{closure } \psi := \forall v_1. \dots \forall v_n. \psi \quad \text{where} \quad FV \psi = \{v_1, \dots, v_n\}$$

Question. Is it necessary to take the closure? Alternatively, for example, can φ range over formulas with v as its only free variable?

Example: every natural number is either even or odd

Deriving $\mathbf{PA} \vdash \forall x. \exists y. (x \equiv y + y \vee x \equiv \text{suc } (y + y))$:



This requires an induction on x . Informally:

- Base case: derive $\text{zero} \equiv \text{zero} + \text{zero}$, which follows from *additionZ* and *symmetry*.
- Inductive case: the induction hypothesis can be analysed into two sub-cases $x' \equiv y' + y'$ and $x' \equiv \text{suc } (y' + y')$.
 - For the first sub-case, derive $\text{suc } x' \equiv \text{suc } (y' + y')$ by *congruence*.
 - For the second sub-case, derive $\text{suc } x' \equiv \text{suc } y' + \text{suc } y'$ by *additionS* and the property below (with the help of *symmetry*, *transitivity*, and *congruence*).

Exercise. Derive $\mathbf{PA} \vdash \forall x. \forall y. \text{suc } (x + y) \equiv x + \text{suc } y$.